

e-CAM82_CUMI0821_MOD

MCU Protocol Application Note



Version 1.6

e-con Systems

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Disclaimer

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Introduction to e-CAM82_CUMI0821_MOD

e-con Systems is a leading Embedded Product Company, which has over two decades of experience in designing, developing, and manufacturing OEM cameras. e-CAM82_CUMI0821_MOD is a new 2-lane MIPI camera which uses the AR0821 camera module.

e-CAM82_CUMI0821_MOD is a low voltage, small form factor, high performance 8 MP pluggable camera module with S-Mount lens holder. It is based on AR0821 CMOS image sensor from onsemi®. e-CAM82_CUMI0821_MOD is designed to connect with any application processor that has MIPI interface. The standard S-Mount lens holder can accommodate a wide range of lenses based on your choice.

This document provides a detailed understanding of I2C based protocol used by the host application processor, for communicating with the microcontroller which is provided as part of e-con Systems e-CAM82_CUMI0821_MOD camera board.

Description

e-CAM82_CUMI0821_MOD can stream uncompressed HD at 30 fps, FHD at 30 fps and 4K at 16 fps UYVY formats. It can be used with any application processor which supports 2-lane MIPI CSI-2 interface.

This document explains the microcontroller unit (MCU) protocol corresponding to the 32-byte MCU firmware version ID **11NL81GMSLXXX01110fd8ae9bXXXXXXXX**.

Note: This MCU firmware version is subject to get updated in the future.

Frame Rate Supported

The below table lists the frame rate supported in e-CAM82_CUMI0821_MOD.

Table 1: Frame Rate Supported

Format	Resolution	DAY HDR (fps)	NIGHT HDR (fps)	Linear (fps)
UYVY	HD	30	60	60
	FHD	30	60	60
	4k	16	16	16

Note: The frame rates listed in the above table varies based on platform capability.

Camera Operation Sequence

This section describes the basic operation for initiating the stream sequence and changing the control values using MCU.

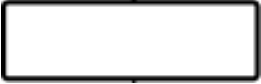
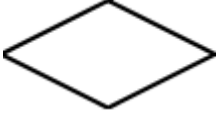
The basic operation for initiating the stream sequence is explained below:

- Host processor refers to the application processor, for example, TX1, TX2, Jetson AGX Orin/Xavier, Xavier NX, TX2-NX and Jetson Nano™ or FX3 processors will act as I²C Master throughout this protocol.
- MCU acts as I²C Slave in this entire protocol.
- The I²C Master always initializes every transaction.
- Length of the byte sequence between the MCU and host processor is either constant or pre-negotiated for each transaction.
- If a transaction is from host processor to MCU, the host processor will perform the following I²C sequence:
 - I²C start condition.
 - 7-bit slave address of MCU.
 - Write bit.
 - Host processor provides data according to the byte sequence defined for that specific command. For more details, please refer to the *MCU Command Description* section.
 - I²C stop condition.
- If a transaction is from MCU to host processor, the host processor will perform the following I²C sequence:
 - I²C start condition.
 - 7-bit slave address of MCU.
 - Read bit.
 - MCU will provide data according to the byte sequence specified in the command.
 - I²C stop condition.
- Checksum is calculated by performing bitwise XOR of the payload data which is not same as the traditional checksum.

The below table lists the legend and its description used in flowchart.

Table 2: Legend and its Description

Legend	Description
	Start or Stop
	MCU Command Block

	Process Block
	Decision Block

Streaming Start Sequence

To start streaming, the sequence to be performed are as follows:

1. Send **Init Camera** command through I²C interface.
2. Send **Get Command Status** command sequence through I²C interface repeatedly, until the return status code is 0x0000.
3. Send **Stream Configure** command with desired frame format, width, height and frame rate.
4. Send **Get Command Status** command sequence through I²C interface repeatedly, until the return status code is 0x0000.
5. Send **Stream ON** command to initiate streaming with updated stream configurations.
6. Send **Get Command Status** command sequence through I²C interface repeatedly, until the return status code is 0x0000.

The flowchart of streaming start sequence is shown below.

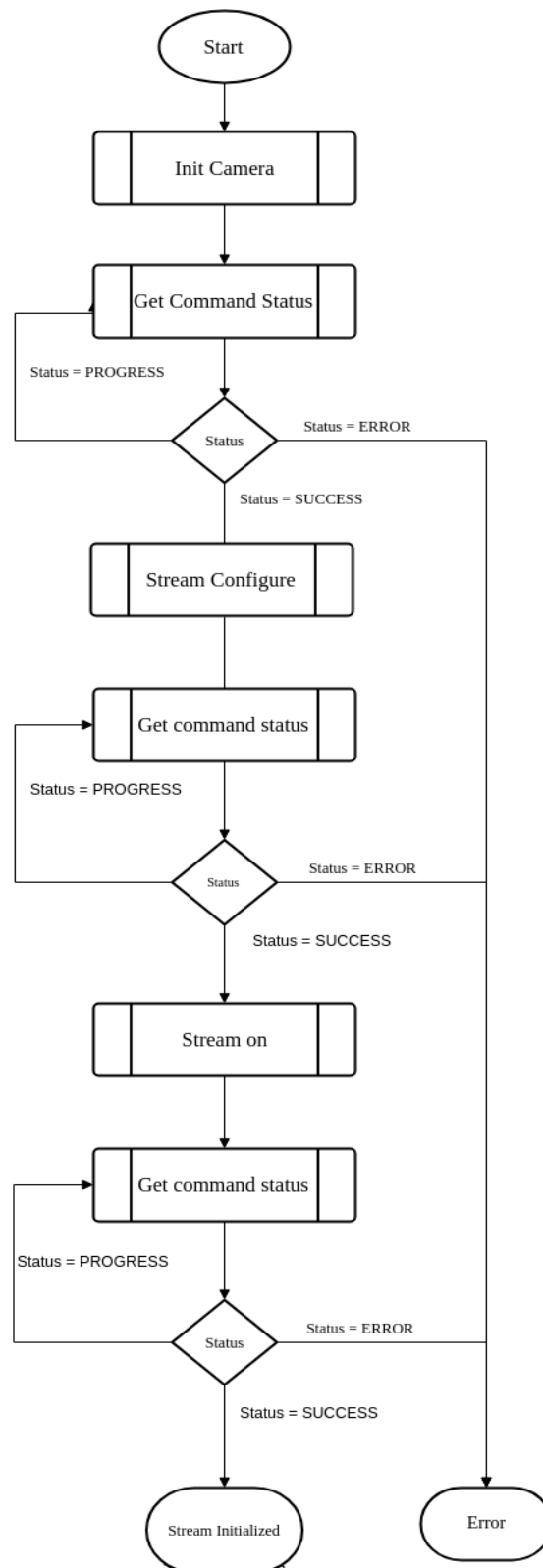


Figure 1: Flowchart of Streaming Start Sequence

Streaming Stop Sequence

To stop streaming, the sequence to be performed are as follows:

1. Send **Stream OFF** command through I²C interface.
2. Send **Get Command Status** command sequence through I²C interface repeatedly, until the return status code is 0x0000.
3. Send **De-Init Camera** command through I²C interface.
4. Send **Get Command Status** command sequence through I²C interface repeatedly, until the return status code is 0x0000.

The flowchart of streaming stop sequence is shown below.

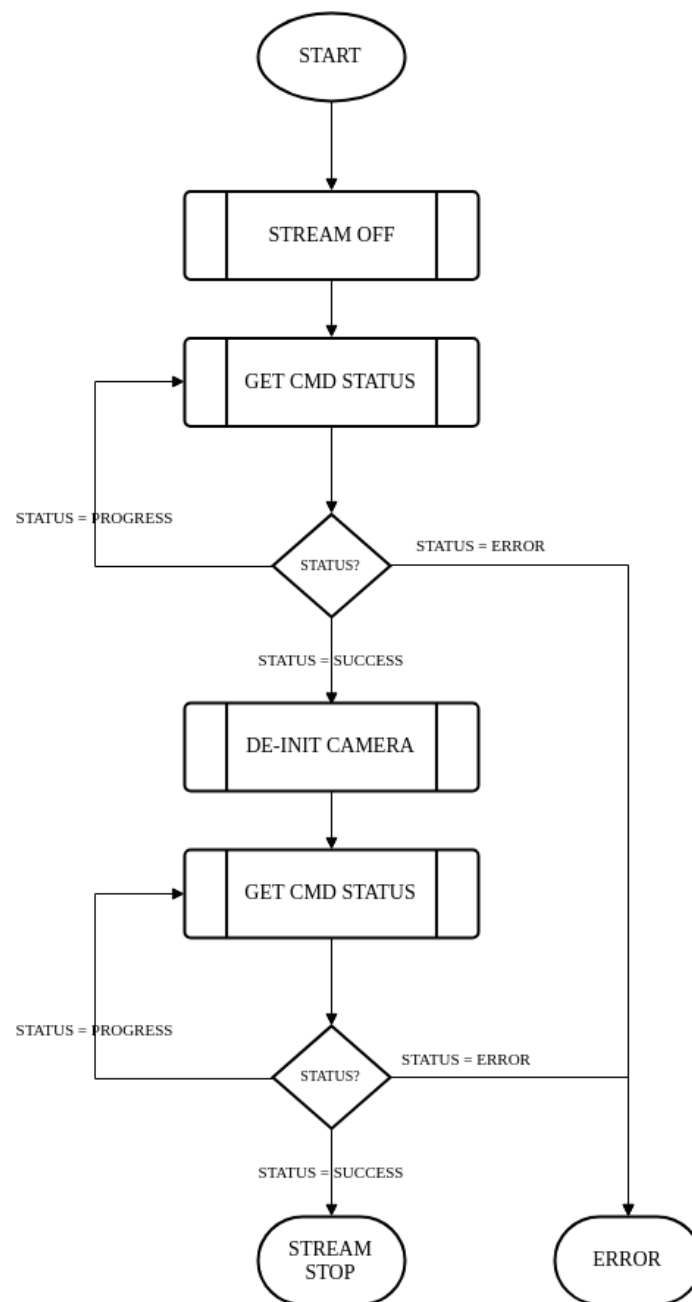


Figure 2: Flowchart of Streaming Stop Sequence

Changing Control Values

To change the control values, the sequence to be followed are as follows:

1. Ensure whether the camera is in streaming state, by performing the sequence of *Streaming Start Sequence* section.
2. Send **Set Control Value** command with desired control index, and ID value.
3. Send **Get Command Status** command sequence through I²C interface repeatedly, until the return status code is 0x0000.

The flowchart of changing control values is shown below.

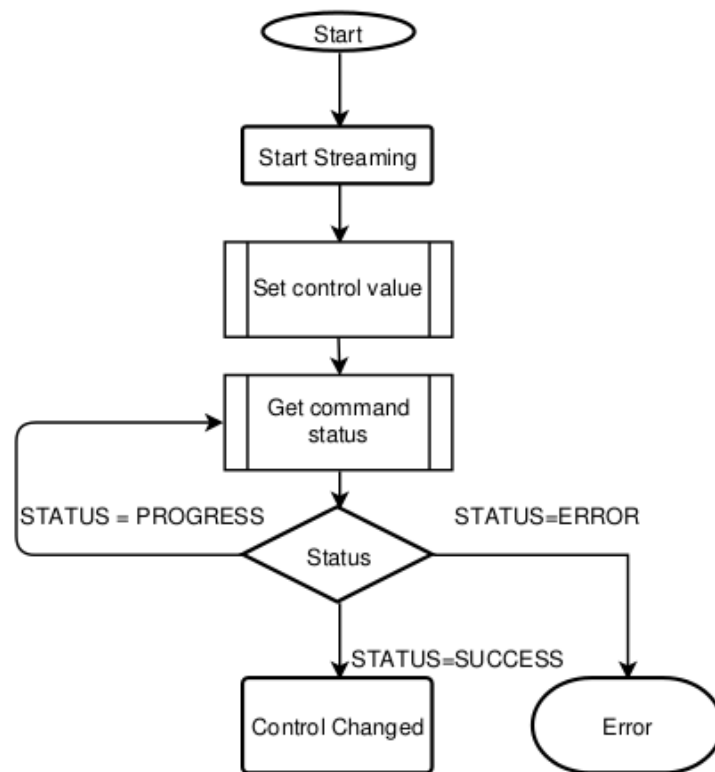


Figure 3: Flowchart of Changing Control Values

Changing Streaming Resolution

To change the streaming resolution, the sequence to be performed are as follows:

1. Follow the steps of *Streaming Stop Sequence* section if the streaming is already started.
2. Follow the steps of *Streaming Start Sequence* section with the desired stream configuration.

MCU Command Overview

This section describes all the commands transferred between the host processor and MCU, and the return values from MCU to Host processor.

MCU I²C Slave Address

MCU has a I²C Slave address of 0x42, which is 7-bit addressing mode.

Note: The above address is required for all the transactions.

Types of Camera Commands

The different types of camera commands are as follows:

- Status command
- Configure command
- Query-Reply command

The below table lists the types of MCU commands.

Table 3: Types of MCU Commands

Status	Configure	Query-Reply
Get Command Status	Init Camera	Get Firmware Version
	De-Init Camera	Get Stream Info
	Stream ON	Get Control Info
	Stream OFF	Get Control Value
	Configure Stream	
	Set Control Value	

List of Camera Commands

The below table lists and describes all the camera commands that are supported by MCU.

Table 4: List of Camera Commands

S.NO	Command	Description	Query Command ID (Hex)
1	Get Firmware Version	This command is used to get the actual firmware version in the MCU.	0x00
2	Init Camera	This command initializes the camera by changing state of hardware pins and writing	0x04

		appropriate settings to the camera. This command returns immediately.	
3	Get Command Status	This command is used to query the status of any executed command. Additionally, it will return the current status of camera (Idle, Busy, and so on) and the hardware specific errors with respect to MCU.	0x05
4	De-Init Camera	This command de-initialize the camera by changing the state of hardware pins and writing the necessary configuration settings.	0x06
5	Stream On	This command will start the camera streaming process.	0x07
6	Stream Off	This command will stop the camera streaming process.	0x08
7	Configure Stream	This command sets the format, width, height and frame rate in the camera. The valid values are specified by Get Stream Info command.	0x09
8	Get Control value	This command gets the value of any control enumerated by Get Control Info command.	0x10
9	Set Control Value	This command sets the value of any control enumerated by Get Control Info command.	0x11

List of Camera Formats

The below table lists the camera format codes that are returned from MCU to host processor.

Table 5: List of Camera Formats

Format Code	Description
0x59565955	YUV 4:2:2 (UYVY)

List of Return Codes

The below table lists all the return codes that are transmitted from MCU to host processor.

Table 6: List of Return Codes

Return Code	Description
0x00	Success or Command Completed

0x01	Busy or Command in Progress
0x02	Invalid Argument
0x03	Permission Denied
0x04	Device Not Found
0x05	I/O Error between ISP and MCU
0x06	Hardware Specific Error
0x07	Try Again
0x08	Already in Effect
0x09	Not Implemented
0x0A	Out of Range
0x0B - 0xFE	Reserved
0xFF	Unknown Failure

Note: For more details, please refer to the *Status Command* section.

List of Command Status Codes

The length of command status is 2-bytes. The command status code is returned by the MCU to the host processor. The below table lists the command status code and its description.

Table 7: List of Command Status Codes (General)

Command Status Code	Description
0x0000	No error or Command Completed
0xF000	Command in Progress

Note: For more details, please refer to the *Status Command* section.

Camera Status Codes

The below table lists the error codes returned by camera to the host processor.

Table 8: List of Command Status Codes (ISP)

Command Status Code	Description
0xFF0	Camera is Powered Down
0xFF1	Camera is Uninitialized

MCU Status Codes

The below table lists the error codes which describes the enumerations of errors specific to MCU.

Table 9: List of Command Status Codes (MCU)

Command Status Code	Description
0x2001	Master I ² C Init Error
0x2002	Master I ² C Timeout
0x2003	Master I ² C I/O Error
0x2004	SPI Init Error
0x2005	SPI Timeout Error
0x2006	SPI I/O Error

0x2007	USART Init Error
0x2008	Framework Error
0x2009	Slave I ² C I/O Error
0x2001	Master I ² C Init Error

Note: The return values help in querying the current state of MCU.

MCU Command Description

This section explains the transactions handled from MCU to host processor while processing the basic MCU Commands.

Status Command

The status command is used to query the status of MCU using the **Get Command Status** command. This command involves three transactions where the reply length from MCU is always constant. The transaction of status command is shown below.

Transaction 1 (Host Processor to MCU) (Write)	0x43	0x05	0x00, 0x01	0x01		
	Communication ID (1 Byte)	Command ID (1 Byte)	Payload Length (2 Bytes)	Check Sum of Payload (1Byte)		
Transaction 2 (Host Processor to MCU) (Write)	0x43	0x05	0xFF			
	Communication ID (1 Byte)	Command ID (1 Byte)	Query Command ID (Payload length)			
Transaction 3 (MCU to Host Processor) (Read)	0x43	0x05	0xFF	0xFF, 0xFF	0xFF	0xFF
	Communication ID (1 Byte)	Command ID (1 Byte)	Command ID of Issued Command (1 Byte)	Command Status Code (2 Bytes)	Check Sum (1 Byte)	Return code (1 Byte)

Figure 4: Transaction of Status Command

Note: The command ID returned by the MCU in Transaction 3 corresponds to the Command ID that was used prior to **Get Command Status** command. Please refer to the *List of Command Status Codes* section to know the various command status codes returned by MCU.

Configure Command

The configure command is used for starting an operation in the sensor through MCU. For example, Stream Config, Set Control Config, Init Cam and so on. This command always returns immediately, while MCU executes the request in the background. The status of the last issued command can be queried through the **Get Command Status** command. The configure command involves two transactions as shown below.

	0x43	0xXX	0xXX, 0xXX	0xXX
Transaction 1 (Write)	Communication ID (1 Byte)	Command ID (1 Byte)	Payload Length (2 Bytes)	Checksum (1 Byte)
	0x43	0xXX	0xXX, 0xXX	0xXX
Transaction 2 (Write)	Communication ID (1 Byte)	Command ID (1 Byte)	Payload Data (Payload Length bytes)	Checksum (1 Byte)

Figure 5: Transaction of Configure Command

The below table lists the transaction values of configure command.

Table 10: Configure Command Transaction Values

Transaction	Packet		Init Camera	De Init Camera	Stream ON	Stream OFF	Configure Stream	Set Control Value
Transaction 1 (Host Processor to MCU)	Communication ID		0x43	0x43	0x43	0x43	0x43	0x43
	Command ID		0x04	0x06	0x07	0x08	0x09	0x11
	Payload Length	Byte 1	0x00	0x00	0x00	0x00	0x00	0xXX, 0xXX (Based on Control Index)
		Byte 2	0x00	0x00	0x00	0x00	0x0E	
	Checksum		0x00	0x00	0x00	0x00	0x0E	0xXX (Based on Control Index)
Transaction 2 (Host Processor to MCU)	Communication ID		0x43	0x43	0x43	0x43	0x43	0x43
	Command ID		0x04	0x06	0x07	0x08	0x09	0x11
	Payload		-	-	-	-	14-byte data (Based on Format Index). For more details, please refer to <i>Table 11</i>	0xXX (Based on Control Index). For more details, please refer to <i>Table 13</i>
	Checksum		-	-	-	-	0xXX (Based on Format Index)	0xXX (Based on Control Type)

Note: Please traverse the above table from top to bottom.

The below table lists the details about the communication to MCU from host, for configure stream.

Table 11: Configure Stream Payload Data

Formats	Payload Data in Bytes						Payload Checksum (1 B)
	Stream Index (2 B)	FourCC Format (4 B)	Width (2 B)	Height (2 B)	Frame Rate (Numerator) (2 B)	Frame Rate (Denominator) (2 B)	
720p at 30 fps (DAY HDR)	0x00, 0x01	0x59,0x56, 0x59,0x55	0x05, 0x00	0x02, 0xD0	0x00, 0x1E	0x00, 0x01	0xCB
720p at 60 fps (LINEAR / NIGHT HDR)	0x00, 0x01	0x59,0x56, 0x59,0x55	0x05, 0x00	0x02, 0xD0	0x00, 0x3C	0x00, 0x01	0xEA
1080p at 30 fps (DAY HDR)	0x00, 0x02	0x59,0x56, 0x59,0x55	0x07, 0x80	0x04, 0x38	0x00, 0x1E	0x00, 0x01	0xA6
1080p at 60 fps (LINEAR / NIGHT HDR)	0x00, 0x02	0x59,0x56, 0x59,0x55	0x07, 0x80	0x04, 0x38	0x00, 0x3C	0x00, 0x01	0x87
4K at 16 fps (HDR/LINEAR)	0x00, 0x03	0x59,0x56, 0x59,0x55	0x0F, 0x00	0x08, 0x70	0x00, 0x16	0x00, 0x01	0x67

Note: Please traverse the above table from left to right.

Query-Reply Command

The query-reply command is used to query information such as Streaming formats, Controls and so on, from MCU. This command uses four transactions between the host and MCU as shown below.

Transaction 1 (Write)	0x43	0xXX	0xXX	0xXX		
	Communication ID (1 Byte)	Command ID (1 Byte)	Payload Length (2 Bytes)	Checksum (1 Byte)		
Transaction 2 (Write)	0x43	0xXX	0xXX, 0xXX		0xXX	
	Communication ID (1 Byte)	Command ID (1 Byte)	Payload Data (Payload Length)		Checksum (1 Byte)	
Transaction 3 (Read)	0x43	0xXX	0xXX, 0xXX	0xXX	0xXX	
	Communication ID (1 Byte)	Command ID (1 Byte)	Reply Length (2 Bytes)	Checksum (1 Byte)	Return Code (1 Byte)	
Transaction 4 (Read)	0x43	0x00	0xXX, 0xXX		0xXX	0xXX
	Communication ID (1 Byte)	Command ID (1 Byte)	Reply Data (Reply Length)		Check sum (1 Byte)	Return code (1 Byte)

Figure 6: Transaction of Query-Reply Command

The below table lists the transaction values of query-reply command.

Table 12: Query-Reply Command Transaction Values

Transaction	Packet		Get Firmware Version	Get Control Value
Transaction 1 (Host Processor to MCU)	Communication ID		0x43	0x43
	Command ID		0x00	0x10
	Payload Length	Byte 1	0x00	0x00
		Byte 2	0x00	0x02
	Checksum		0x00	0x02
Transaction 2 (Host Processor to MCU)	Communication ID		0x43	0x43
	Command ID		0x00	0x10
	Payload Data		-	0xXX, 0xXX (Control Index)
	Checksum		-	0x00
Transaction 3 (MCU to Host Processor)	Communication ID		0x43	0x43
	Command ID		0x00	0x10
	Reply Length	Byte 1	0x00	0x00
		Byte 2	0x20	0x09
	Checksum		0x20	0x09
	Return Code		0x00	0x00
Transaction 4 (MCU to Host Processor)	Communication ID		0x43	0x43
	Command ID		0x00	0x10
	Reply Data		Firmware version - 32-bytes	Based on Control Index. For more details, please refer to <i>Table 13</i>
	Checksum		Based on reply data	Based on reply data
	Return Code		0x00	0x00

Note: Please traverse the above table from top to bottom.

Controls of e-CAM82_CUMI0821_MOD

This section describes the controls available in e-CAM82_CUMI0821_MOD, which can be set or get through the MCU.

The available controls of e-CAM82_CUMI0821_MOD are as follows:

- Brightness
- Contrast
- Saturation
- White Balance (both manual and automatic)
- Gamma
- Gain
- Horizontal Flip
- Vertical Flip
- Sharpness
- Exposure (auto, manual and region of interest (ROI) based exposure)
- ROI Window Size
- HDR
- Trigger
- Frame Sync
- Trigger mode
- Denoise
- Exposure Compensation
- Power line frequency selection
- Frame Rate Control
- Special Effects

The below table lists the get or set control payload data of e-CAM82_CUMI0821_MOD controls.

Table 13: Get or Set Control Payload Data

Control Name	Control Index (2 B)		Payload Data in Bytes								
			Control ID (4 B)				Control Type (1 B)	Current Value (4 B)			
Brightness	0x00	0x00	0x00	0x98	0x09	0x00	0x01	0xFF	0xFF	0xFF	0xFF
Contrast	0x00	0x01	0x00	0x98	0x09	0x01	0x01	0xFF	0xFF	0xFF	0xFF
Saturation	0x00	0x02	0x00	0x98	0x09	0x02	0x01	0xFF	0xFF	0xFF	0xFF
White Balance (Auto)	0x00	0x03	0x00	0x98	0x09	0x0C	0x01	0xFF	0xFF	0xFF	0xFF
Gamma	0x00	0x04	0x00	0x98	0x09	0x10	0x01	0xFF	0xFF	0xFF	0xFF
Gain	0x00	0x05	0x00	0x98	0x09	0x13	0x01	0xFF	0xFF	0xFF	0xFF
Horizontal Flip	0x00	0x06	0x00	0x98	0x09	0x14	0x01	0xFF	0xFF	0xFF	0xFF
Vertical Flip	0x00	0x07	0x00	0x98	0x09	0x15	0x01	0xFF	0xFF	0xFF	0xFF

White Balance Temperature	0x00	0x08	0x00	0x98	0x09	0x1a	0x01	0xXX	0xXX	0xXX	0xXX
Sharpness	0x00	0x09	0x00	0x98	0x09	0x1b	0x01	0xXX	0xXX	0xXX	0xXX
Exposure Selection	0x00	0x0A	0x00	0x9A	0x09	0x01	0x01	0xXX	0xXX	0xXX	0xXX
Exposure Time, Absolute	0x00	0x0B	0x00	0x9A	0x09	0x02	0x01	0xXX	0xXX	0xXX	0xXX
ROI Window Size	0x00	0x0C	0x00	0x9A	0x09	0x24	0x01	0xXX	0xXX	0xXX	0xXX
HDR	0x00	0x0D	0x00	0x9A	0x09	0x28	0x01	0xXX	0xXX	0xXX	0xXX
TRIGGER	0x00	0x0E	0x00	0x9A	0x09	0x29	0x01	0xXX	0xXX	0xXX	0xXX
Frame Sync	0x00	0x0F	0x00	0x9A	0x09	0x2B	0x01	0xXX	0xXX	0xXX	0xXX
Trigger Mode	0x00	0x10	0x00	0x9A	0x09	0x2C	0x01	0xXX	0xXX	0xXX	0xXX
Denoise	0x00	0x11	0x00	0x9A	0x09	0x2D	0x01	0xXX	0xXX	0xXX	0xXX
Exposure Compensation	0x00	0x12	0x00	0x9A	0x09	0x31	0x01	0xXX	0xXX	0xXX	0xXX
Powerline frequency selection	0x00	0x13	0x00	0x9A	0x09	0x32	0x01	0xXX	0xXX	0xXX	0xXX
Frame Rate Control	0x00	0x14	0x00	0x9A	0x09	0x29	0x01	0xXX	0xXX	0xXX	0xXX
Special Effects	0x0	0x15	0x00	0x9A	0x09	0x2A	0x01	0xXX	0xXX	0xXX	0xXX

Note: Please traverse the above table from left to right.

Brightness Control

The brightness values can be changed from a minimum value of -15 to 15. This control increases the low light performance of e-CAM82_CUMI0821_MOD. The default value is 0.

Range: -15 (0xFFFFFFFF1) to 15 (0x0000000F)

Data Type: 32-bit signed integer

Control ID: 0x00980900

Setting (Write) Brightness Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Brightness Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Contrast Control

The contrast values can be changed from a minimum value of 0 to 10. Increasing the contrast value increases the luminance of e-CAM82_CUMI0821_MOD. The default value is 5.

Range: 0 (0x00000000) to 10 (0x0000000A)

Data Type: 32-bit unsigned integer

Control ID: 0x00980901

Setting (Write) Contrast Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Contrast Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Saturation Control

The saturation values can be changed from a minimum value of 0 to 60. Increasing the saturation value increases the intensity of the color of e-CAM82_CUMI0821_MOD. The default value is 16.

Range: 0 (0x00000000) to 60 (0x0000003C)

Data Type: 32-bit unsigned integer

Control ID: 0x00980902

Setting (Write) Saturation Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Saturation Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Auto White Balance Control

The white balance values can be toggled between 0 (manual) to 1 (auto). The default value is 1.

Range: 0 (0x00000000) to 1 (0x00000001)

Data Type: 32-bit unsigned integer

Control ID: 0x0098090C

Setting (Write) White Balance Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of *Table 10*.

Getting (Read) Current White Balance Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of *Table 12*.

White Balance Temperature Control

The manual white balance values can be changed from 10 to 10000. The default value is 4500. This control can be enabled only when auto white balance is set to 0 (manual).

Range: 10 (0x0000000A) to 10000 (0x00002710)

Data Type: 32-bit unsigned integer

Control ID: 0x0098091A

Setting (Write) White Balance Temperature Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of *Table 10*.

Getting (Read) Current White Balance Temperature Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of *Table 12*.

Gamma Control

The gamma values can be changed from a minimum value of 40 to 500. The default value is 220.

Range: 40 (0x00000028) to 500 (0x000000DC)

Data Type: 32-bit unsigned integer

Control ID: 0x00980910

Setting (Write) Gamma Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of *Table 10*.

Getting (Read) Current Gamma Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of *Table 12*.

Gain Control

The gain values can be changed from a minimum value of 1 to 100. The changes will be updated only when exposure mode is set to manual. The default value is 1.

Range: 1 (0x00000001) to 100 (0x00000064)

Data Type: 32-bit unsigned integer

Control ID: 0x00980913

Setting (Write) Gain Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Gain Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Horizontal Flip Control

The preview from the sensor can be horizontally flipped by setting the horizontal flip mode control value to 1 (ON). The default value is 0 (OFF).

Values: 0x00000000 (OFF) or 0x00000001 (ON)

Data Type: 32-bit unsigned integer

Control ID: 0x00980914

Setting (Write) Horizontal Flip Value

MCU Command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Horizontal Flip Value

MCU Command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Vertical Flip Control

The preview from the sensor can be vertically flipped by setting the vertical flip mode control value to 1 (ON). The default value is 0 (OFF).

Values:

Data Type: 32-bit unsigned integer

Control ID: 0x00980915

Setting (Write) Vertical Flip Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Vertical Flip Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Sharpness Control

The sharpness values can be changed from a minimum value of 0 to 7. This control increases the image clarity of e-CAM82_CUMI0821_MOD. The default value is 2.

Range: 0x00000000 to 0x00000007

Data Type: 32-bit unsigned integer

Control ID: 0x0098091B

Setting (Write) Sharpness Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Sharpness Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Exposure Control

e-CAM82_CUMI0821_MOD supports manual, auto and ROI based exposure controls. When exposure time is changed, the frame rate varies accordingly. The default exposure mode is Full FOV Auto Mode.

Values: 0x00000000 (Full FOV Auto Mode)

0x00000001 (Manual Mode)

0x00000002 (ROI based Auto Mode)

Data Type: 32-bit unsigned integer

Control ID: 0x009A0901

Setting (Write) Exposure Control Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Exposure Control Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Exposure Time, Absolute (Manual Exposure) Control

The manual exposure control is used to set the absolute exposure time for e-CAM82_CUMI0821_MOD. When exposure time is changed, the frame rate varies accordingly. The values range from 1 to 10000. The default value is 312. This control can be changed only when exposure control is set to manual mode.

Values: 1 (0x00000001) to 10000 (0x00002710)

Data Type: 32-bit unsigned integer

Control ID: 0x009A0902

Setting (Write) Manual Exposure Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Manual Exposure Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

ROI Window Size Control

The ROI window size can be changed from a minimum value of 8 to 64. The default size is 8. This control can only be changed in steps of 8, for example, 8, 16, 24 and so on.

Values: 8 (0x00000008) to 64 (0x00000040)

Data Type: 32-bit unsigned integer

Control ID: 0x009A0924

Setting (Write) ROI Window Size Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current ROI Window Size Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

HDR

The boot mode control is used to operate the sensor in HDR or Linear mode. The default mode is HDR.

Values: 0x00000000 (Day HDR) or 0x00000001 (Night HDR) or 0x00000010 (Linear)

Data Type: 32-bit unsigned integer

Control ID: 0x009A0928

Setting (Write) Boot mode Control Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Boot mode Control Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Trigger

e-CAM82_CUMI0821_MOD supports this control to toggle between internal and external mode of trigger.

Values: 0x00000000 (Internal_Trigger) or 0x00000001 (External_Trigger)

Data Type: 32-bit unsigned integer

Control ID: 0x009A0929

Setting (Write) Trigger Value

MCU Command type: Configuration command. You can refer to the Set Control Value of Table 10.

Getting (Read) Trigger Value

MCU Command type: Query-reply command. You can refer to the Get Control Value of Table 12.

Frame Sync Control

e-CAM82_CUMI0821_MOD provides frame sync control to switch between master and trigger modes. When switched to trigger mode, the camera will provide frame synchronous to an external trigger signal. The default mode is master.

Values: 0x00000000 (Frame Sync 15 Hz)

0x00000001 (Frame Sync 30 Hz)

0x00000010 (Frame Sync 60 Hz)

Data Type: 32-bit unsigned integer

Control ID: 0x009A092B

Setting (Write) Frame Sync Value

MCU Command type: Configuration command. You can refer to the *Set Control Value* of Table 10.

Getting (Read) Frame Sync Value

MCU Command type: Query-reply command. You can refer to the *Get Control Value* of Table 12.

Trigger Mode

e-CAM82_CUMI0821_MOD provides trigger mode control to switch between free running and trigger only modes. The default mode is free running mode, in this mode camera frames are sync to trigger only when trigger signal is detected, otherwise run freely. When switched to trigger only mode, the camera outputs the frame only when trigger signal is detected. In Disabled mode the camera provides frame continuously.

Values: 0x00000000 (Disabled Mode)

0x00000001 (Free Running Mode)

0x00000010 (Trigger Only Mode)

Data Type: 32-bit unsigned integer

Control ID: 0x009A092C

Setting (Write) Trigger Mode Value

MCU Command type: Configuration command. You can refer to the *Set Control Value* of Table 10.

Getting (Read) Trigger Mode Value

MCU Command type: Query-reply command. You can refer to the *Get Control Value* of Table 12.

Denoise Control

The denoise control is used to reduce noise in low lighting conditions. The default value is 8.

Values: 0 (0x00000000) to 15 (0x0000000F)

Data Type: 32-bit unsigned integer

Control ID: 0x009A092E

Setting (Write) Denoise Control Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of *Table 10*.

Getting (Read) Current Denoise Control Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of *Table 12*.

Exposure Compensation Control

The exposure compensation control adjusts the upper limit of auto exposure. The default value is 33333.

Values: 8000 (0x00001F40) to 1000000 (0x000F4240)

Data Type: 32-bit unsigned integer

Control ID: 0x009A0932

Setting (Write) Exposure Compensation Control Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of *Table 10*.

Getting (Read) Current Exposure Compensation Control Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of *Table 12*.

Getting (Read) Current Power Line Frequency Control Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of *Table 12*.

Power Line Frequency Selection Control

Power line frequency is used to select the anti-damping frequency when sensor is exposed to artificial light sources such as tube lights and computer screens. The default value is auto.

Values: 0x00000000 (auto mode), 0x00000001 (50HZ mode) or 0x00000002 (60HZ mode).

Data Type: 32-bit unsigned integer

Control ID: 0x009A0932

Setting (Write) Power Line Frequency Control Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Frame Rate Control

Frame rate control is used to control the required frame rate of the camera. The default value is 30.

Range: 0 (0x00000000) to 60 (0x0000003C)

Data Type: 32-bit signed integer

Control ID: 0x00980929

Setting (Write) Frame Rate Control Value

MCU command type: Configuration command. Please refer to the *Set Control Value* of Table 10.

Getting (Read) Current Frame Rate Control Value

MCU command type: Query-Reply command. Please refer to the *Get Control Value* of Table 12.

Special Effects

The special effects enable four other processed image preview options.

Values: 0x00000000 (Normal)

0x00000001 (Black and White)

0x00000010 (GrayScale)

0x00000011 (Negative)

0x00000100 (Sketch).

Data Type: 32-bit unsigned integer

Control ID: 0x009A092A

Revision History

Rev	Date	Description	Author
1.0	22-Aug-2022	Initial draft	Camera Dev Team
1.1	30-Jan-2023	MCU firmware version and Exposure compensation value updated	Camera Dev Team
1.2	10-Feb-2023	Added Sync Support	Camera Dev Team
1.3	22-Mar-2023	Updated MCU Firmware version	Camera Dev Team
1.4	24-May-2023	Updated MCU Firmware Version and Frame Rate	Camera Dev Team
1.5	07-Jun-2023	Updated MCU Firmware Version	Camera Dev Team
1,6	08-Jan-2024	Updated MCU Firmware Version	Camera Dev Team